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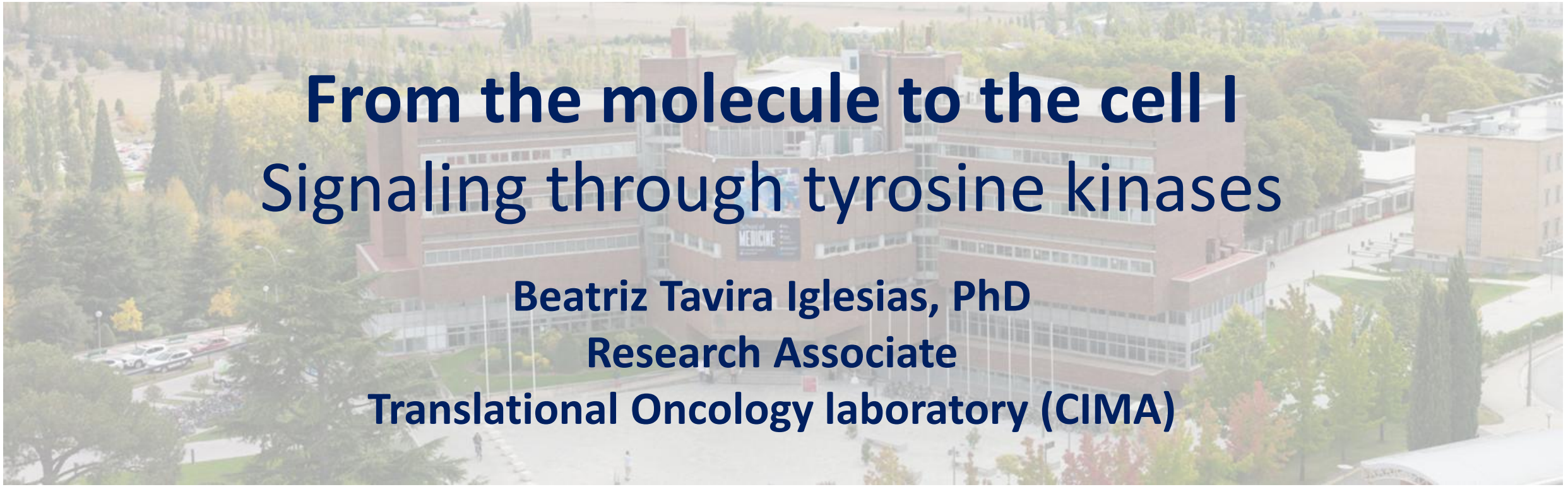
Class instructions

Please, sit together according the current working groups

Write the **group´s name** in the first page.

Unnamed writing tasks- will NOT be evaluated (score=0)

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From the molecule to the cell I

Signaling through tyrosine kinases

Beatriz Tavira Iglesias, PhD
Research Associate
Translational Oncology laboratory (CIMA)

What are going to learn today…

1-Introduction to signaling membrane receptors.

2- What is an enzymatic or catalytic receptor : tyrosine-kinase receptors (TKR).

3- Medical relevance of TKR:

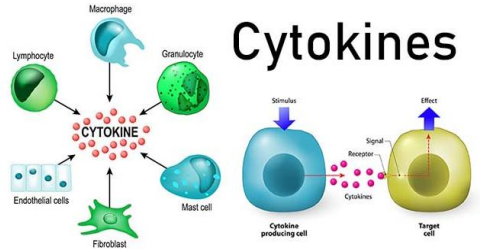
- **Cancer –EGFR (1st test)**
  - **10 min break**

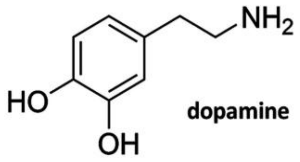
4-Writing task by groups : finishing clinical case EGFR, 2nd test and clinical case of diabetes

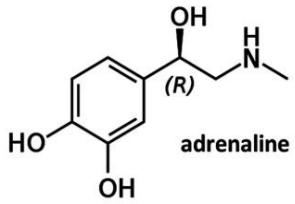
Introduction to signaling membrane receptors.

Cellular communication is...

diverse







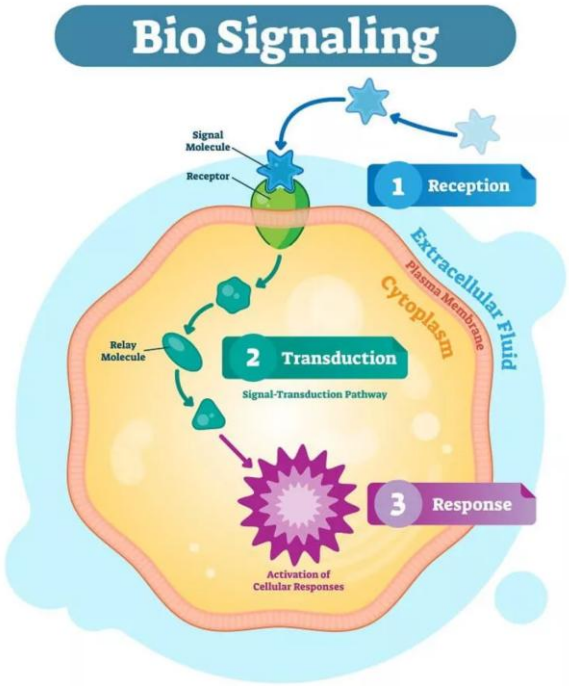
Hormones

IGF-1

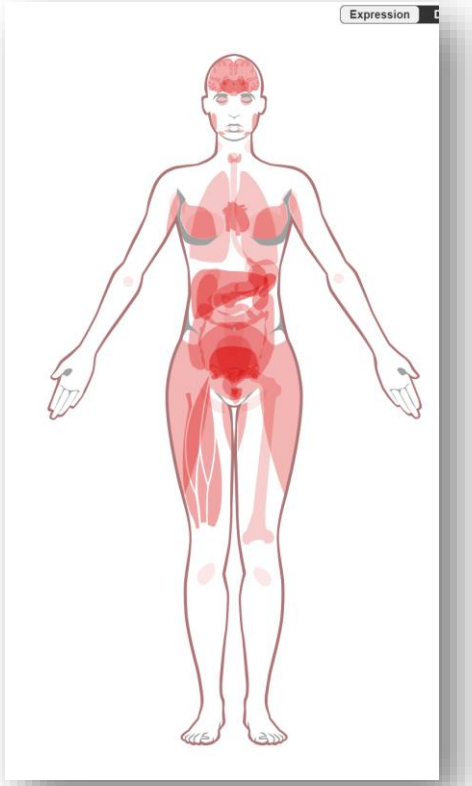
FGF VEGF

Factors

needed

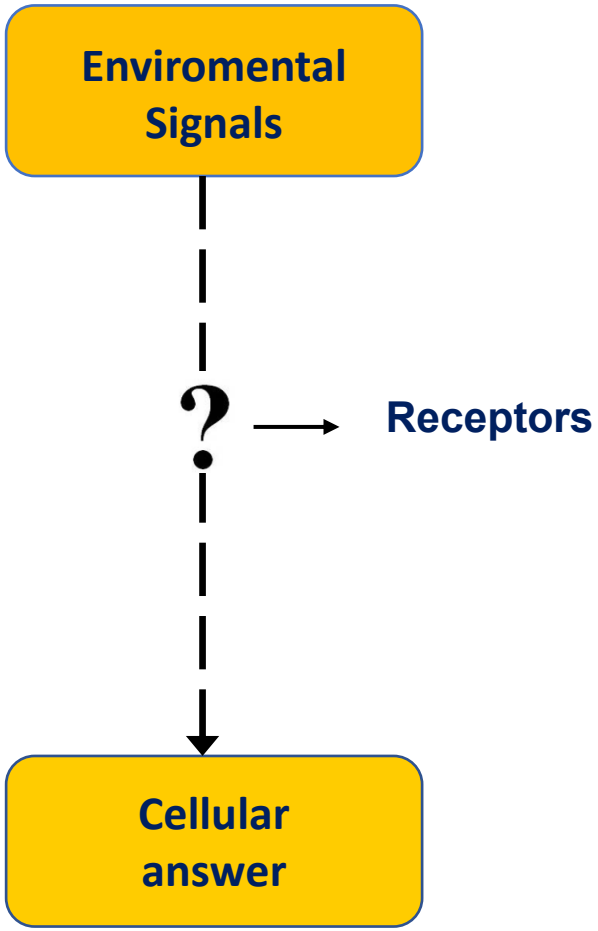


Human body works

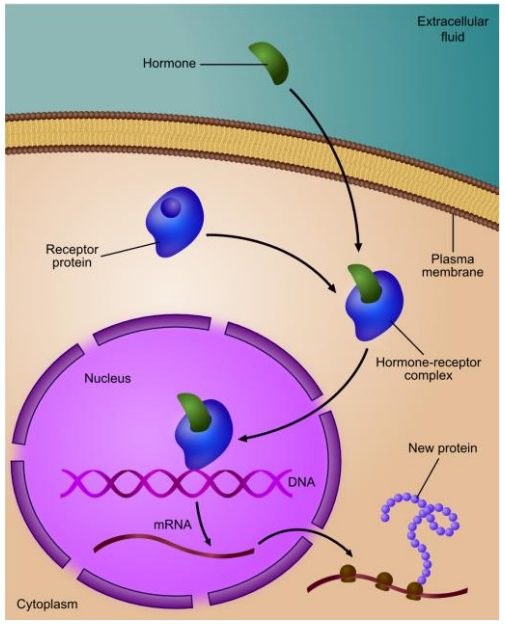


Introduction to signaling membrane receptors.

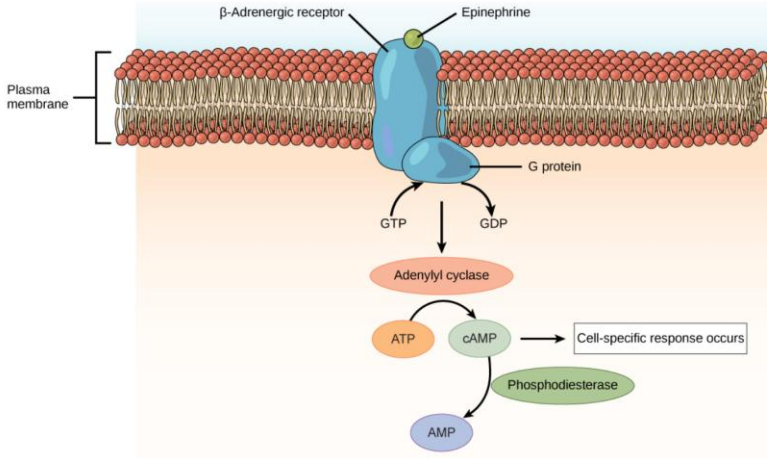
Communication is possible due to...



1. Collection extracellular signals→ transmission interior of the cell
2. Activation intracellular cascade → gene transcription →protein synthesis.



Intracellular receptors (hormones)



Membrane receptors

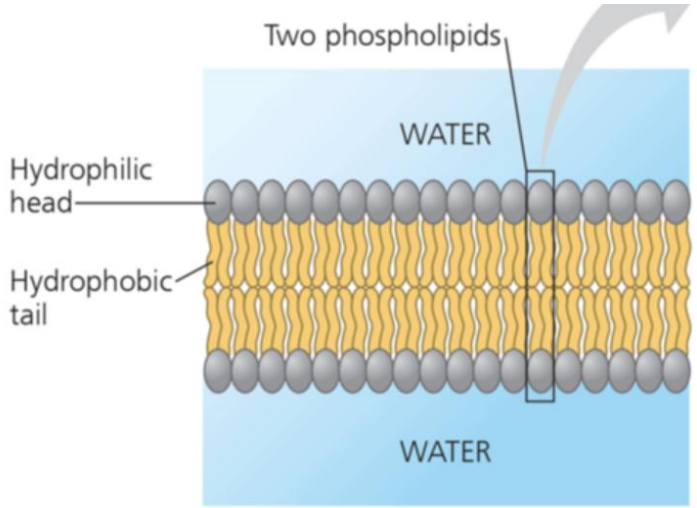
Ali Zifan 03:07, 10 July 2016 (UTC), CC BY-SA 4.0 <https://creativecommons.org/licenses/by-sa/4.0>, via Wikimedia Commons

Introduction to signaling membrane receptors.

Signaling through membrane receptors

What are membrane receptors?

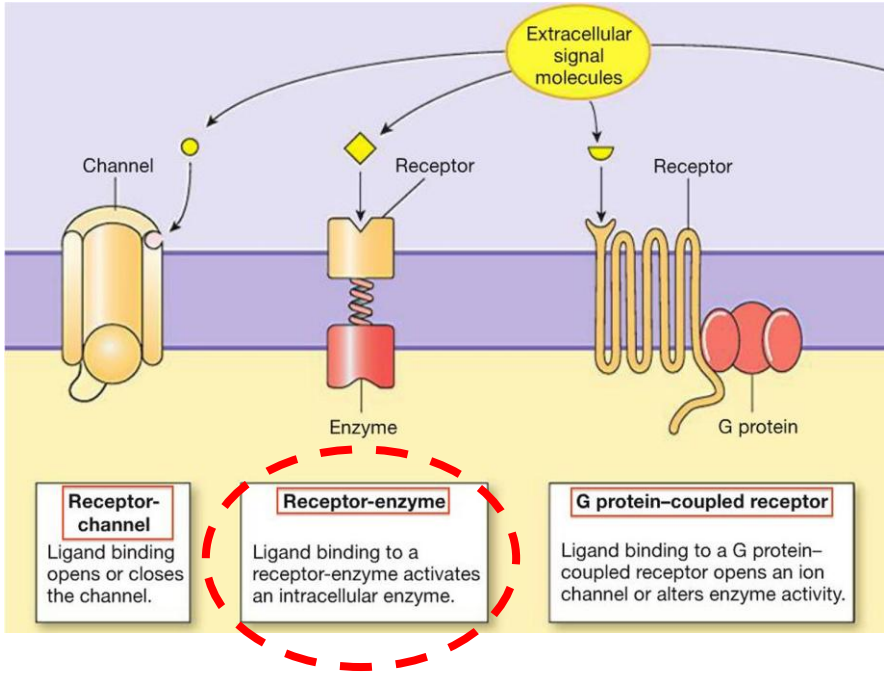
1. Located in the membrane
2. Amphipathic molecules
3. Signal transduction→ gene activation→protein synthesis.



Membrane receptors types?

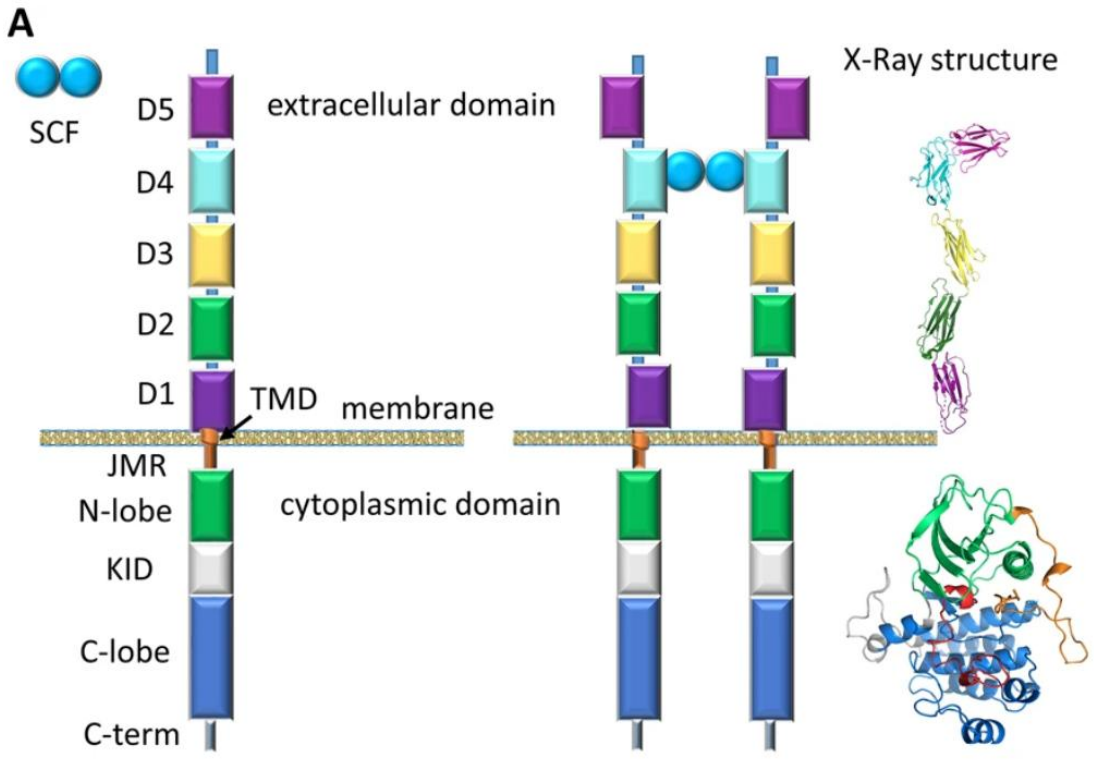
3 main groups:

- Ligand gated ion channels
- G-protein coupled receptors
- Enzymatic or catalytic



What is an enzymatic or catalytic receptor : tyrosine-kinase receptors (TKR).

Enzymatic or catalytic receptors



Extracellular domain→NH2 terminal→
ligand binding

Transmembrane domain

Intracellular domain →**catalytic activity**
kinase activity

What is a kinase?
What is kinase-activity?

?

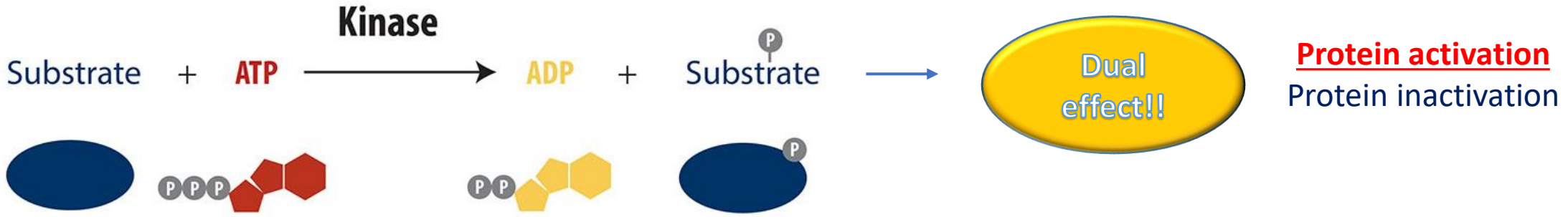
What is an enzymatic or catalytic receptor

Kinase activity

Kinases: proteins/enzymes that catalizes the transfer of phosphate groups from a donor **ATP molecule** **to a substrate***.

Protein phosphorylation by kinase enzymes regulates countless different cellular signaling pathways.

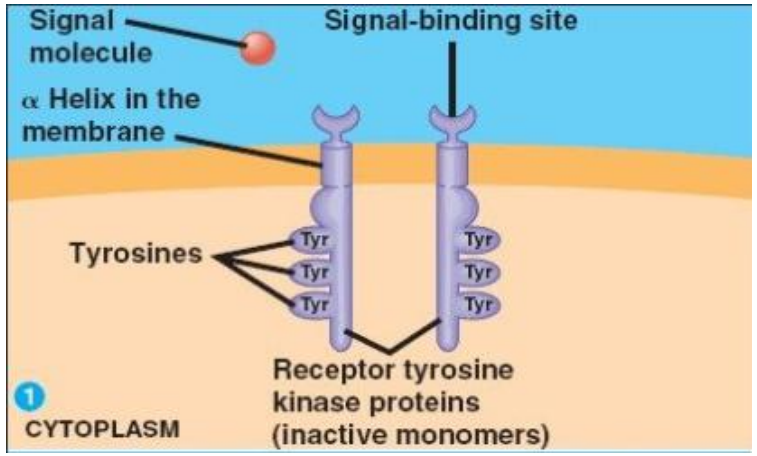
*substrates: aminoacids, proteins, lipids, carbohydrates and nucleic acids.



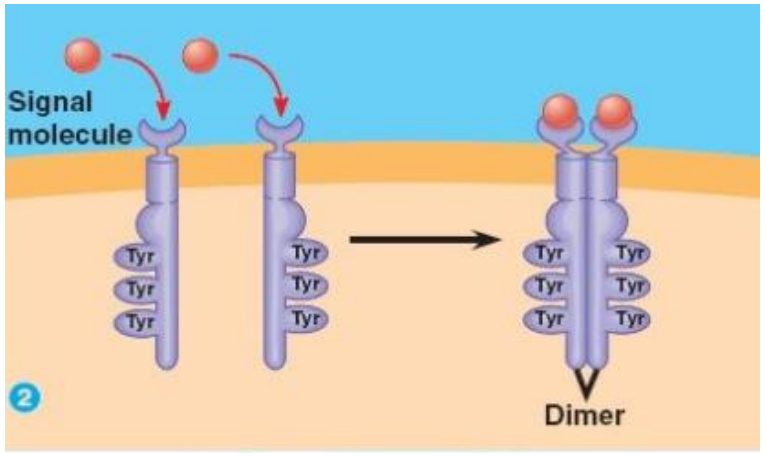
Video: https://www.youtube.com/watch?v=xG2WOd_fWqo

What is an enzymatic or catalytic receptor

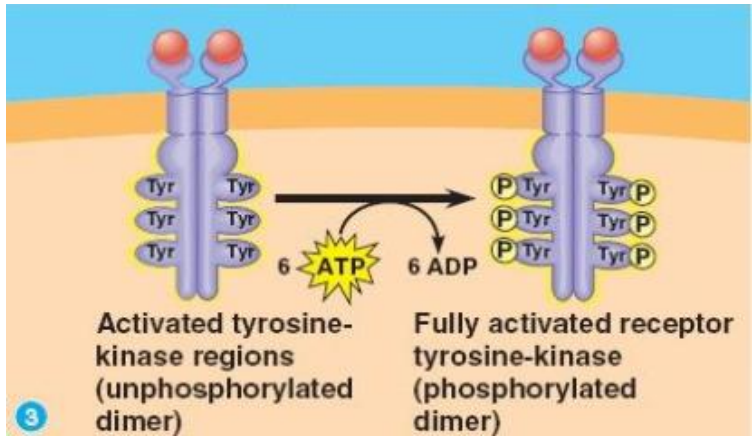
TKR -how do they work??



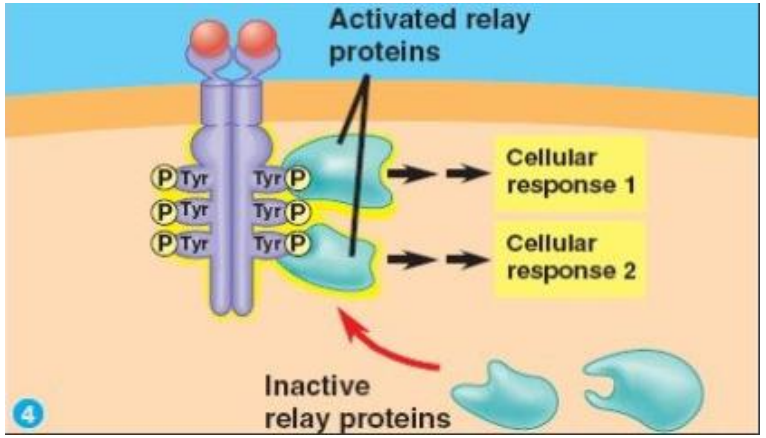
Inactive form:
monomeric



Preactivation form:
Dimer no phosphorylated



Active form:
Dimer phosphorylated (crossphosphorylation)

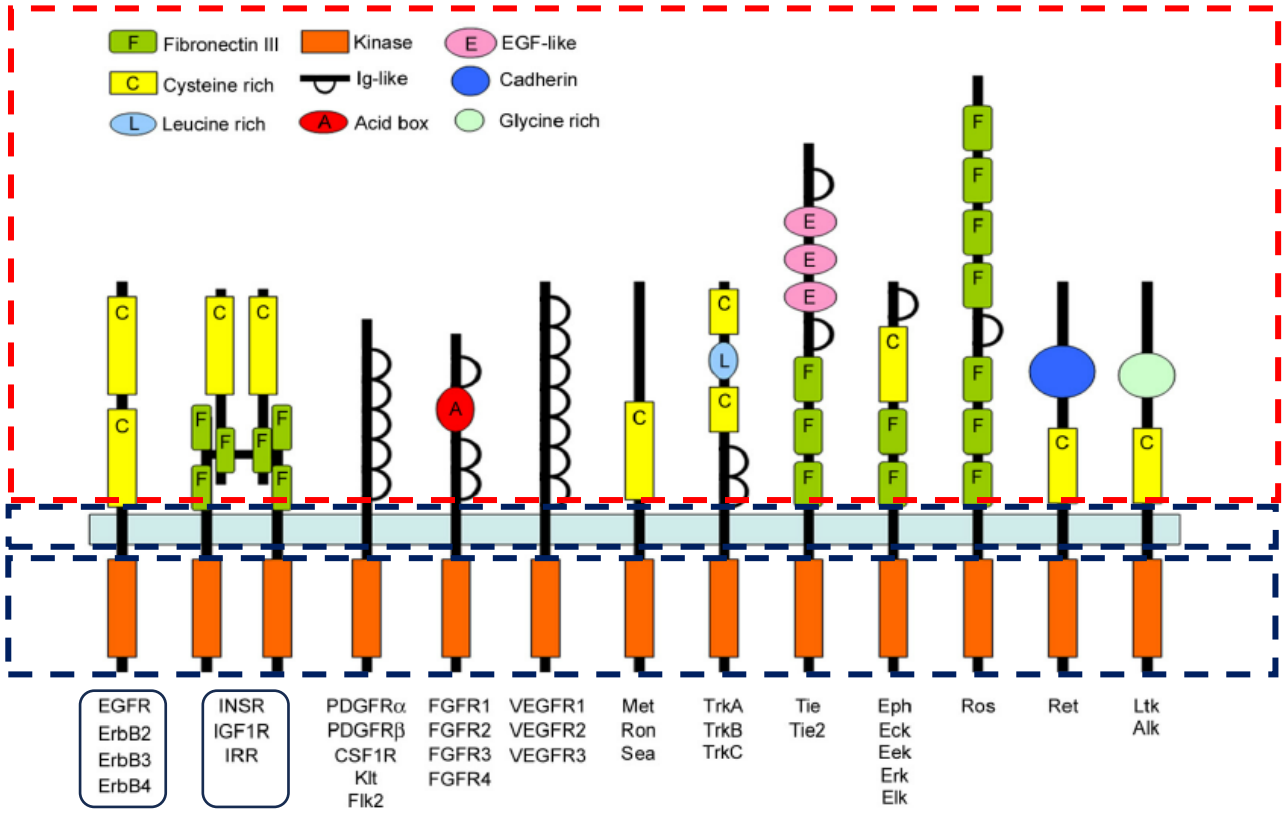


Active form:

- Atracction different accesory proteins.
- Downstream protein activation
- Negative feedback-
- Stop activation

What is an enzymatic or catalytic receptor

TKR family



Specificity!

Extracellular domain

Transmembrane domain

Intracellular domain

What is an enzymatic or catalytic receptor

TKR : tip of the iceberg

Tyrosin-kinase receptors



Tyrosin-kinase receptors



Intracellular signaling activated by tyrosin-kinase receptors

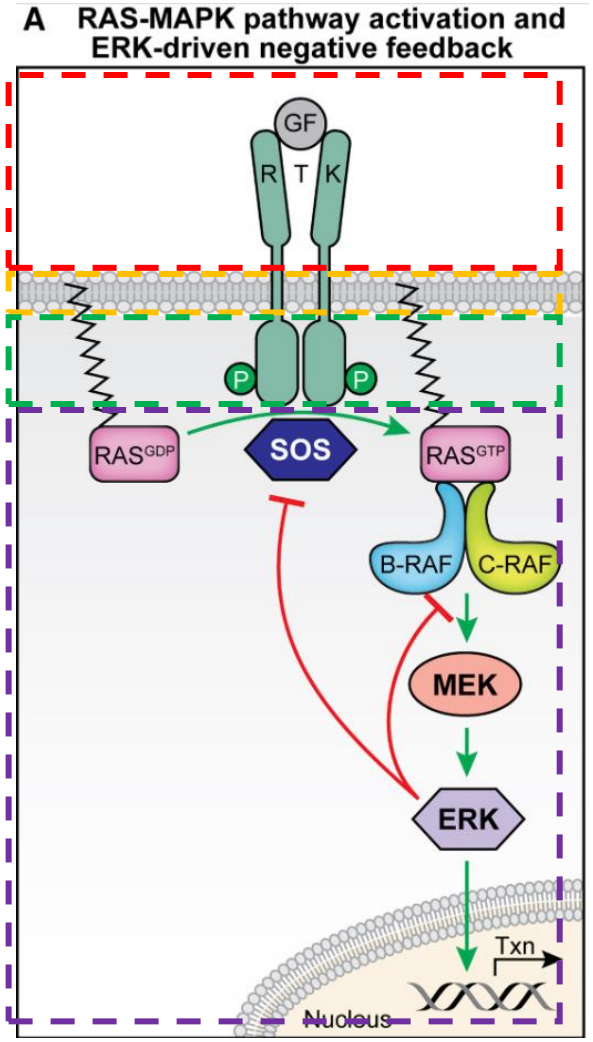
Extracellular domain

Transmembrane domain

Intracellular domain

Intracellular signaling

Medical relevance



Medical relevance of TKR

TKR are …



Eg: INSR distribution

- Widely distributed in the body

Regulate many developmental, metabolic and physiological processes:

- cell survival
- cell growth
- cell division

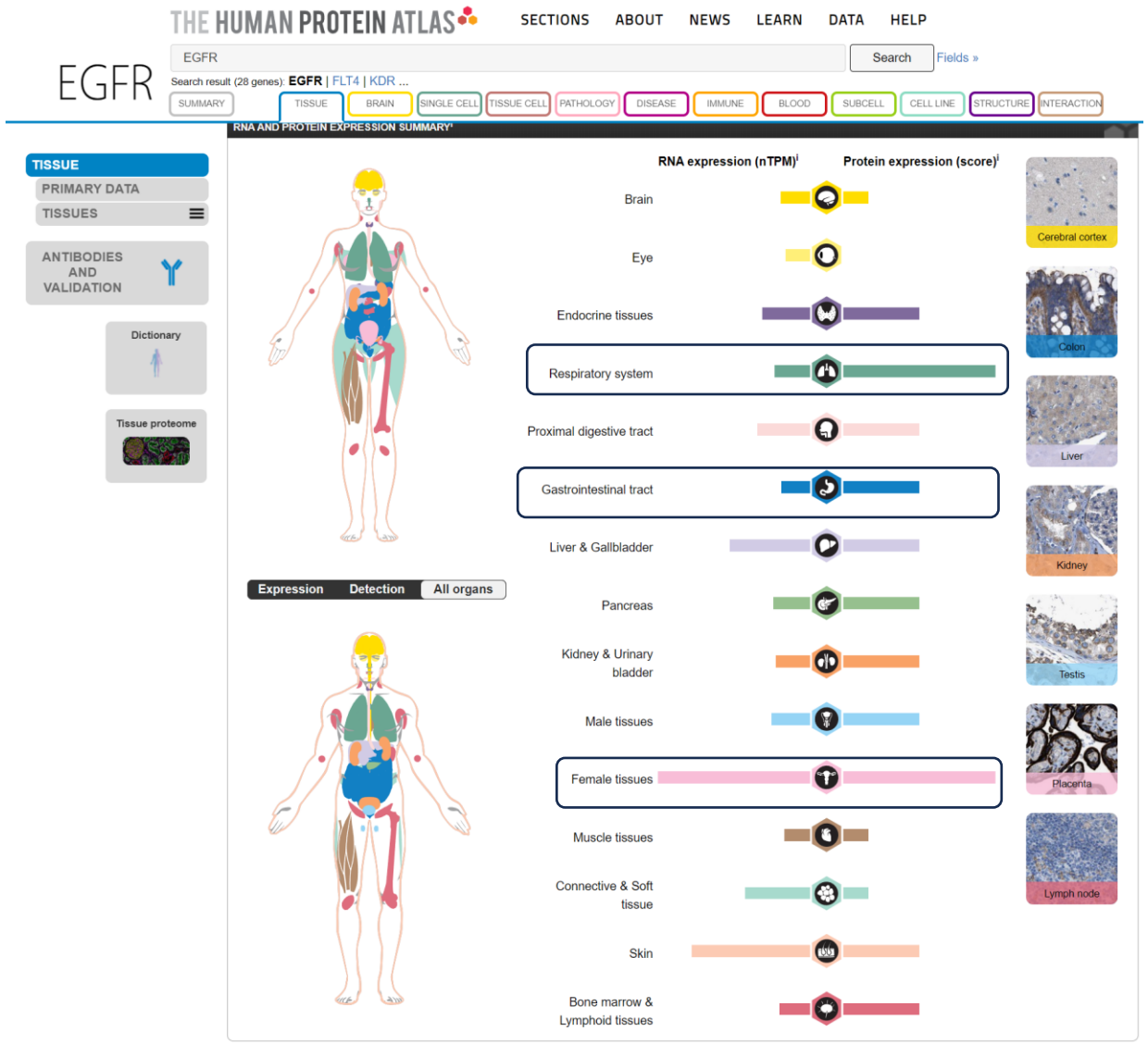
Epidermal Growth Factor Receptor (EGFR) family

Insulin receptor (INSR) family

Involved in disease development: Cancer/diabetes

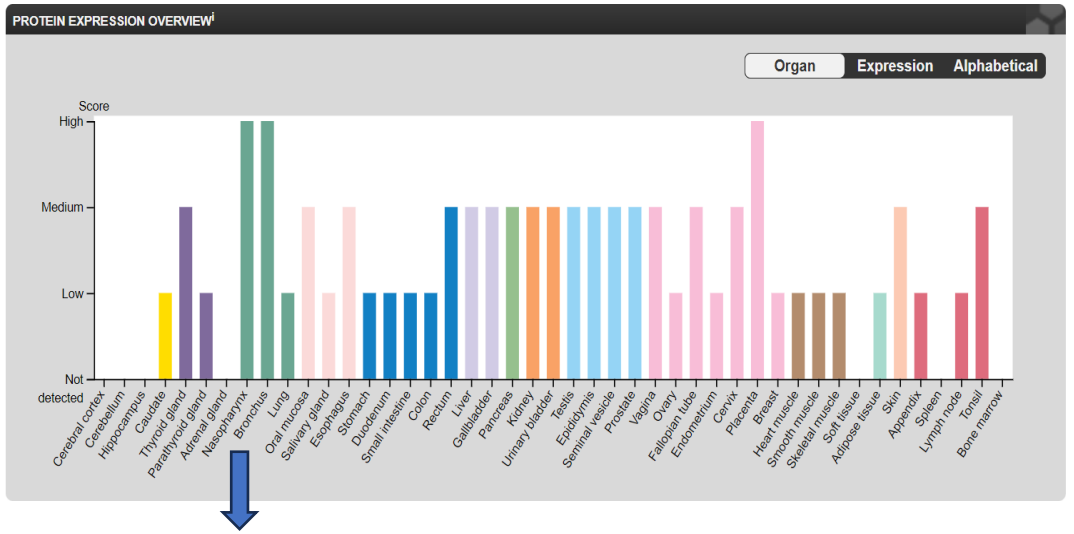
Medical relevance of TKR in cancer

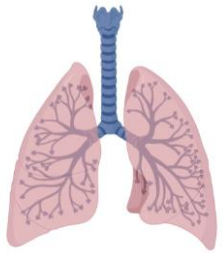
TKR: EGFR family



Epidermal Growth Factor Receptor (EGFR) family : wide expression different organs

Dysregulation is involved in cancer development





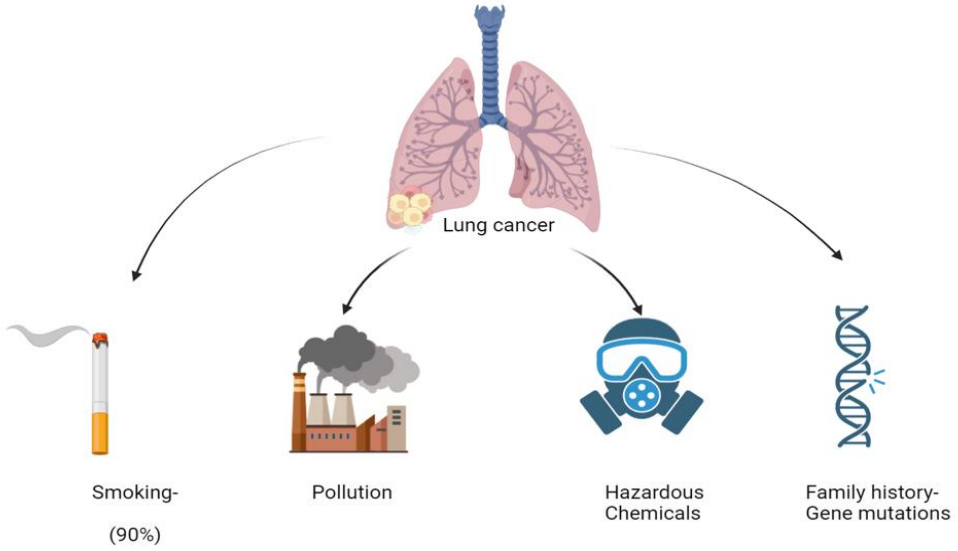
Lung cancer

Medical relevance of TKR in cancer

TKR : EGFR family

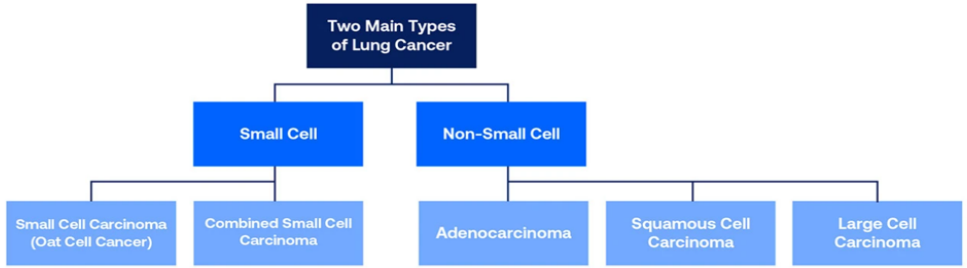
Lung cancer: 1st cause of death for cancer in Spain
20% deaths for cancer

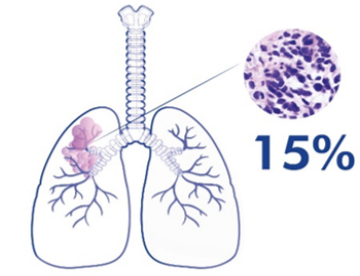
Multifactorial disease.
Factors risk

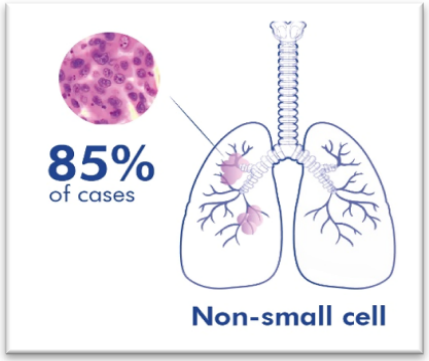


What Are the Types of Lung Cancer?

There are two main types of lung cancer: small cell lung cancer (SCLC) and non-small cell lung cancer (NSCLC). A third less common type of lung cancer is called carcinoid.





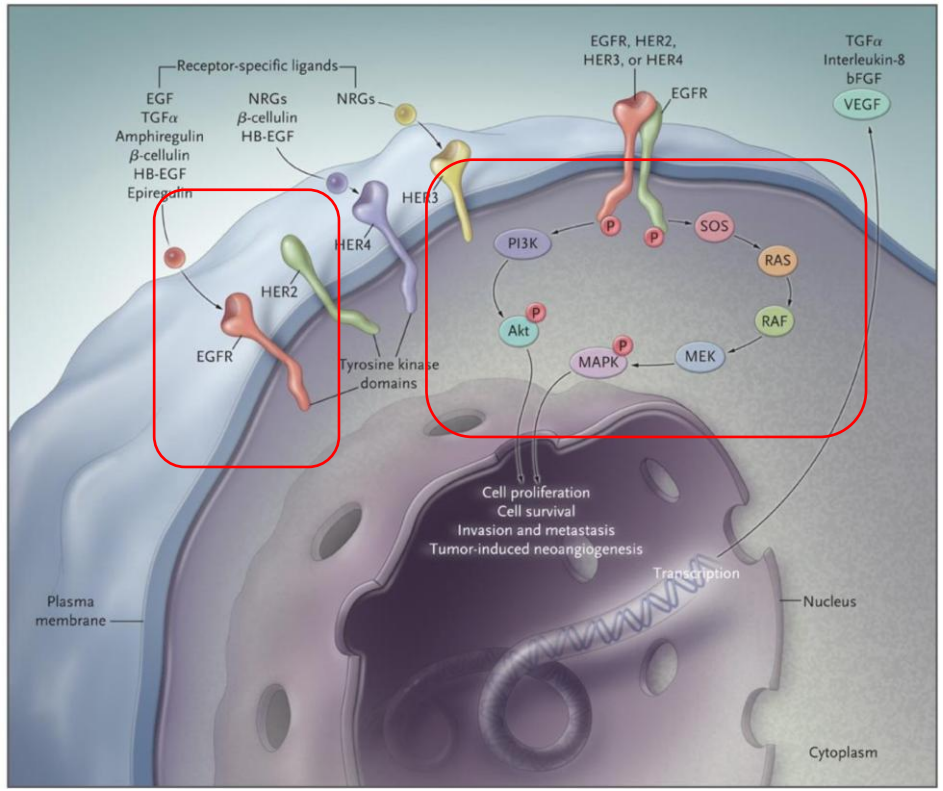


5-year relative survival rate~ 28%

Epidermal Growth Factor Receptor (EGFR) and lung cancer… why is so important?

Medical relevance of TKR in cancer

TKR : EGFR family

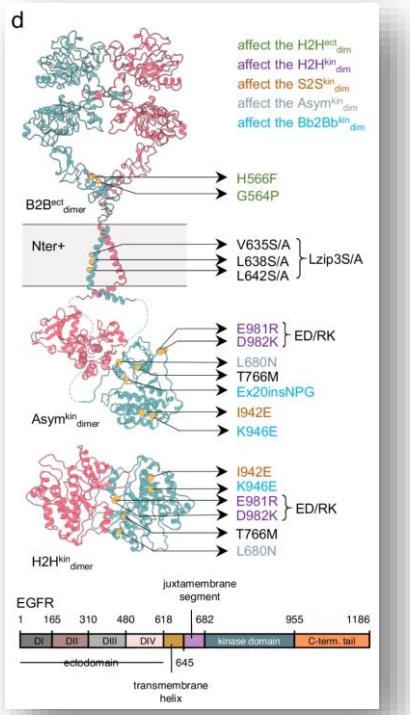


1-Ligand+ EGFR: canonical activation of the pathway. Mutations in EGFR increase the kinase activity-> activation pathways downstream.

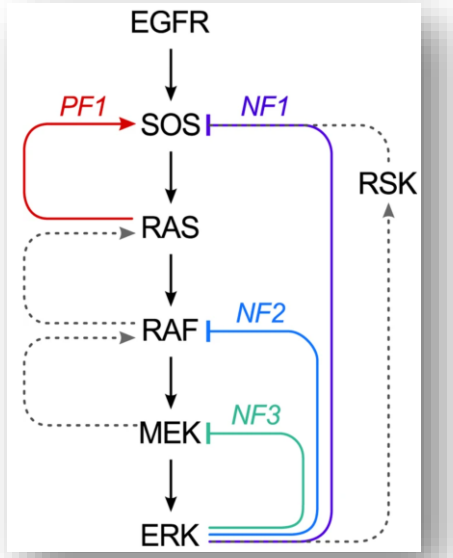
2-No ligand+EFGR : The mutation promotes kinase domain activity and reduces dependence on the normal stimulus. Proliferative signaling can therefore persist even in the absence of the ligand.

- Cell proliferation
- Cell survival
- Invasion
- Metastasis
- Neoangiogenesis

1-EGFR mutations



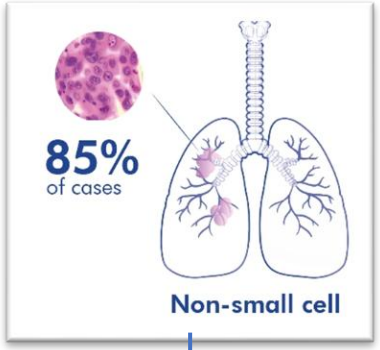
2- Functional receptor but dysregulation RAS-MAPK signaling pathway- RAS mutations



Ciardiello F, Tortora G. EGFR antagonists in cancer treatment. N Engl J Med. 2008 Mar 13;358(11):1160-74. doi: 10.1056/NEJMra0707704. Erratum in: N Engl J Med. 2009 Apr 9;360(15):1579. PMID: 18337605.

Medical relevance of TKR in cancer

TKR : 1-Mutations in EGFR

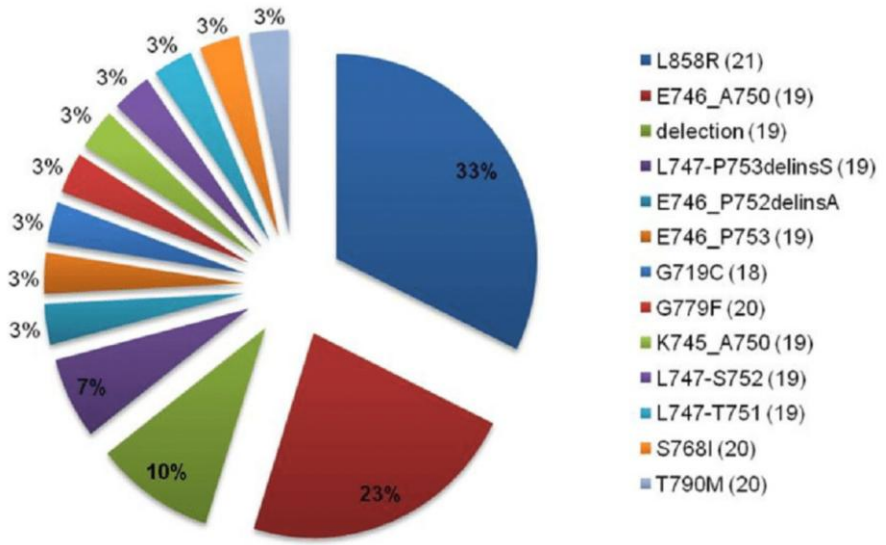


Activating mutations in the kinase domain of EGFR.

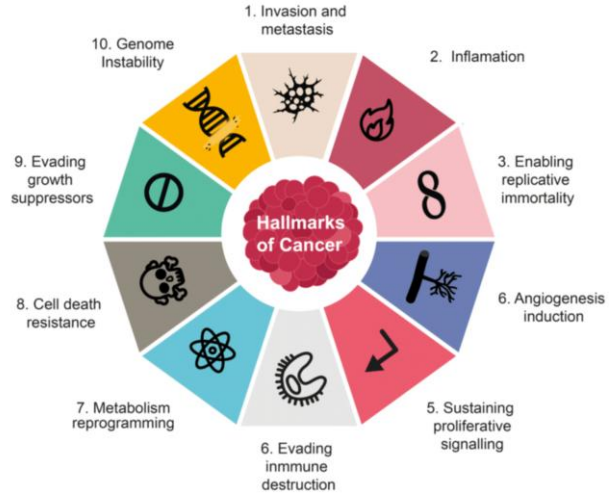


↑ **a net increase in kinase activity**

EGFR mutations

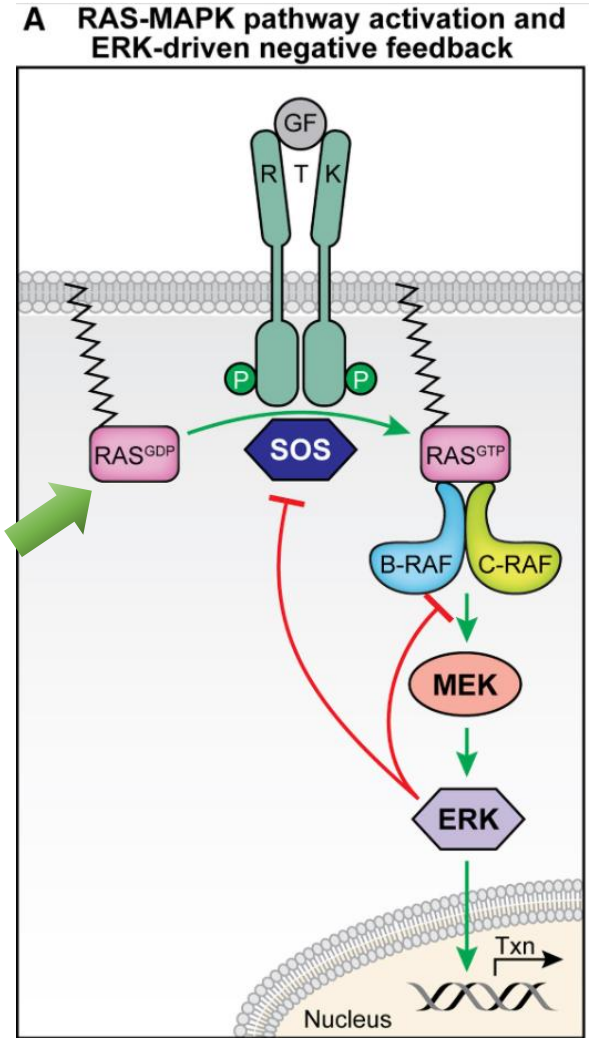


↑ Tumoral phenotype
Cell proliferation, migration,metastasis



Medical relevance of TKR in cancer

TKR : 2-Dysregulation RAS-MAPK signaling pathway- RAS mutations.



RAS protein is involved in :

- cytoskeletal reorganization
- cell polarity
- cell cycle progression
- gene expression the interaction with foreign particles.

3 Isoforms: KRAS, NRAS and HRAS.

Dysregulation of RAS→ RAS mutations→Activating mutations (OFF→ON)



↑ **a net increase RAS activity-signaling transmission**

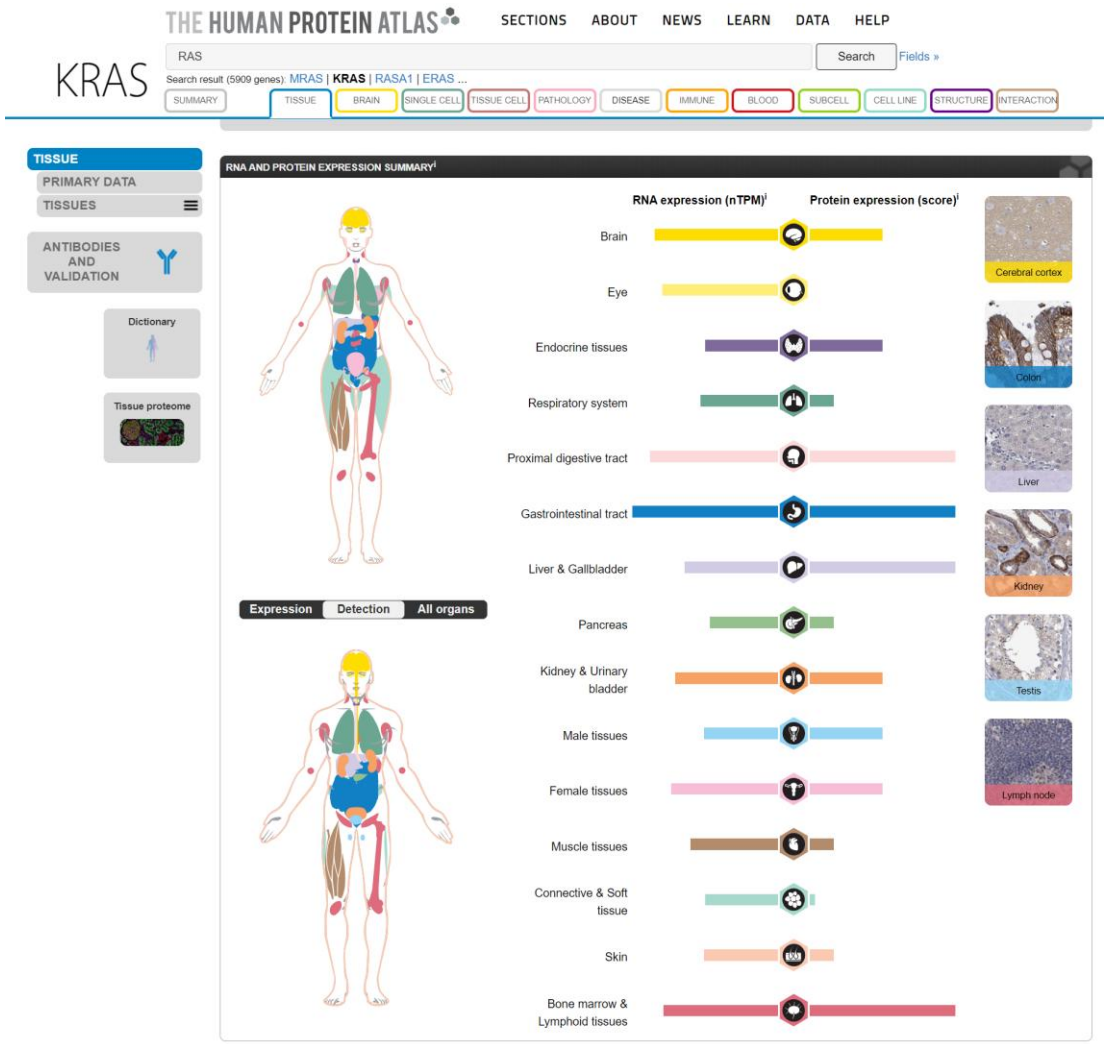
Medical relevance

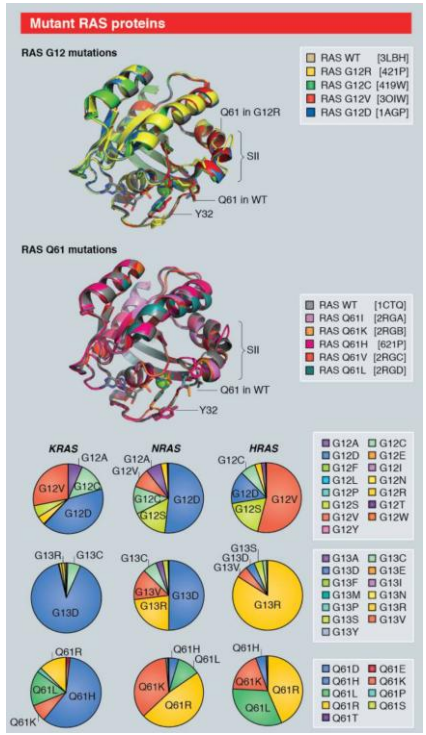
RAS mutations

- pancreas (90%)
- lung (35%)
- thyroid gland (55%)
- colon (45%)
- liver (30%)

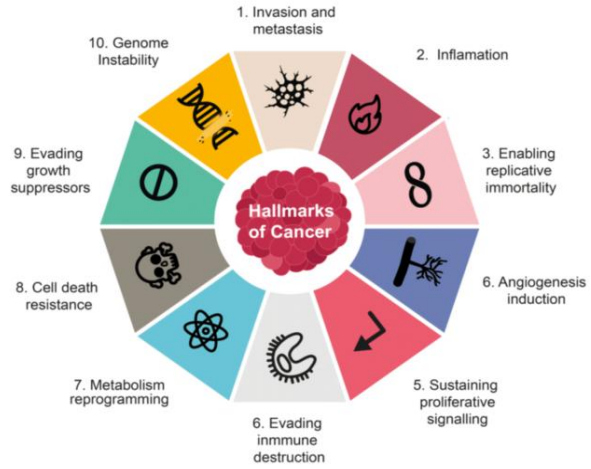
Medical relevance of TKR in cancer

TKR : Mutations in RAS protein





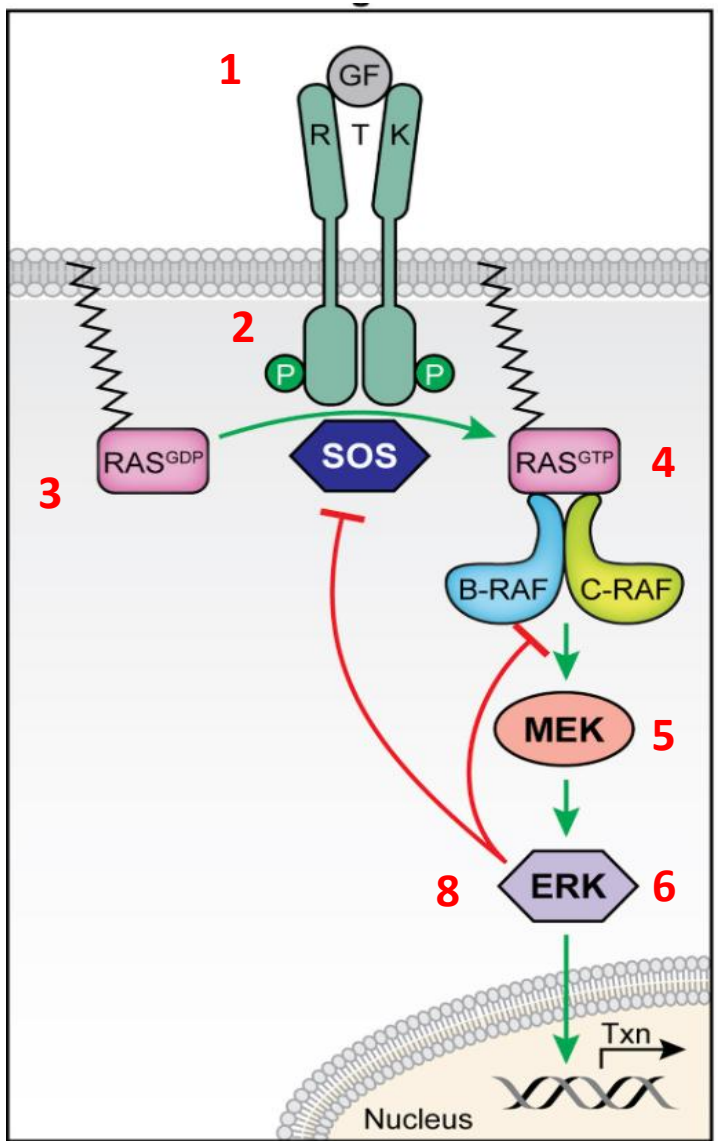
Tumoral phenotype
Cell proliferation, migration,metastasis



Medical relevance of TKR in cancer

Intracellular signaling activated by TKR : Ras-MAPK signaling

Ras-MAPK signaling



Also known as **RAS-RAF-MEK-ERK** pathway.

1- Ligand binds to membrane receptor / EGR→ EGFR

2- Tyrosin kinase receptor activation/ Crossphosphorilation

3- Activation of RAS from "dormant" state (GDP→ GTP)

4- Activation of RAF isoforms

5- Activation of MEK

6- Activation of ERK (MAPK)

7- Nuclei translation of ERK→ gene activation→ protein synthesis

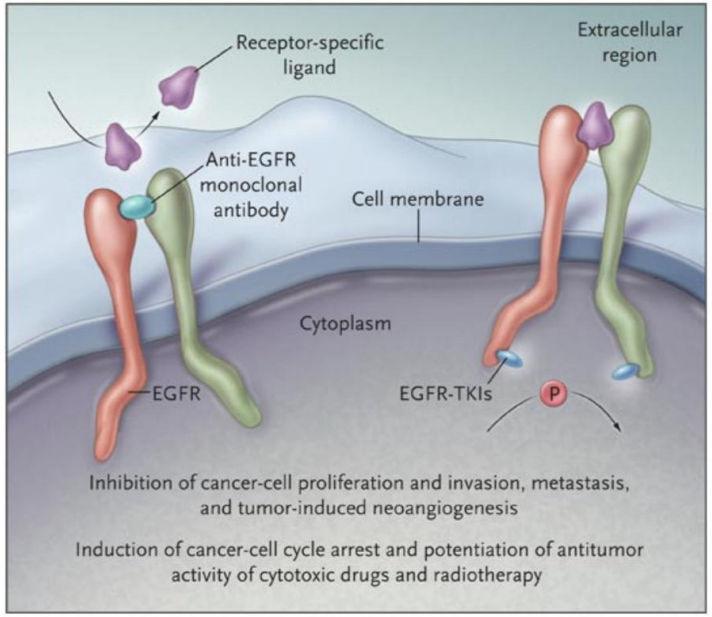
8- Negative feedback (control protein synthesis, cell division)

https://www.youtube.com/watch?v=ql62kMK3BrM

Applications of TKR drugs in cancer

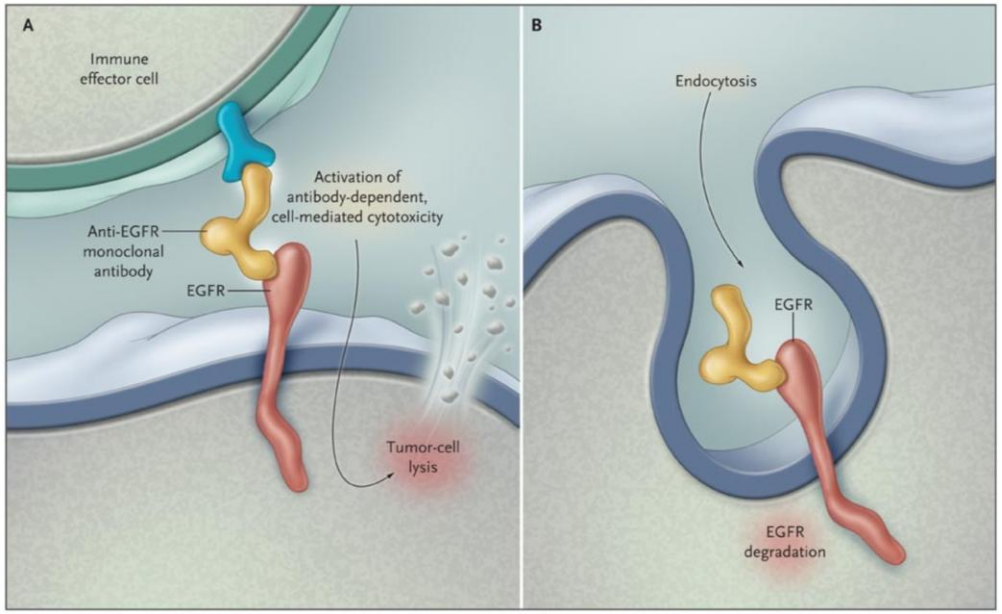
As TKRs play an important role in cancer development→ potential target for cancer therapy

Strategies for tackling cancer through
Epidermal Growth Factor Receptor (EGFR)- Hyperactivation of receptor



Blocking ligand receptor binding

Blocking ligand ATP substrate (erlotinib)



Anti-EGFR monoclonal antibody (cetuximab)

Limited efficacy- resistance development

Ciardiello F, Tortora G. EGFR antagonists in cancer treatment. N Engl J Med. 2008 Mar 13;358(11):1160-74. doi: 10.1056/NEJMra0707704. Erratum in: N Engl J Med. 2009 Apr 9;360(15):1579. PMID: 18337605.

Applications of TKR drugs in cancer

Do NOT study



Strategies for tackling cancer through Epidermal Growth Factor Receptor (EGFR)

Table 2. EGFR Inhibitors Currently Approved for Cancer Treatment.*

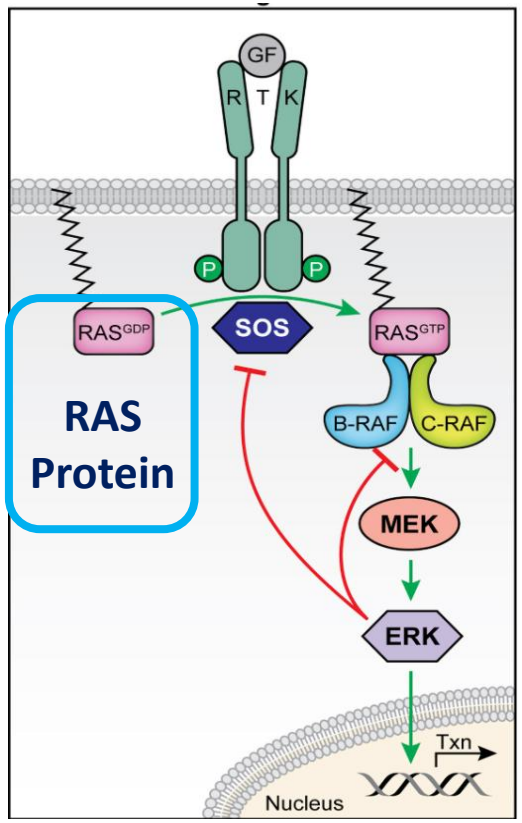
Drug	Molecular Properties	Approved Uses
Erlotinib	Reversible EGFR tyrosine kinase inhibitor (quinazoline-derivative molecule)	Erlotinib has been approved by several regulatory agencies worldwide, including the FDA and the EMEA in the European Union, as monotherapy for the treatment of non–small-cell lung cancer that is refractory to platinum-based chemotherapy. More recently, erlotinib has been approved by the FDA and the EMEA for use in combination with gemcitabine as first-line treatment for advanced pancreatic cancer.
Gefitinib	Reversible EGFR tyrosine kinase inhibitor (quinazoline-derivative molecule)	Gefitinib has been approved in various countries for use as third-line treatment of non–small-cell lung cancer that is refractory to platinum-based and docetaxel-based chemotherapy regimens. After an accelerated approval process, it was approved by the FDA in May 2003 but has been withheld from the U.S. market since June 2005, as a result of the release of preliminary results of the ISEL trial, which assessed its use in patients with non–small-cell lung cancer that was refractory to previous platinum-based chemotherapy. Gefitinib has never been approved in the European Union but is currently on the market in Japan, Korea, China, and several other Asian countries. It is currently an investigational drug in the United States and the European Union.
Cetuximab	Human–mouse chimeric monoclonal antibody (IgG1 subtype)	Cetuximab has been approved by several regulatory agencies worldwide, including the FDA and the EMEA, for the treatment of advanced colorectal cancer that is refractory to irinotecan-based chemotherapy (alone or in combination with irinotecan in the United States but only in combination with irinotecan in the European Union). Cetuximab in combination with radiotherapy is also approved for the treatment of locally advanced squamous-cell carcinoma of the head and neck.
Panitumumab	Fully human monoclonal antibody (IgG2κ subtype)	Panitumumab has been approved by several regulatory agencies worldwide, including the FDA, as monotherapy for third-line treatment of colorectal cancer that is refractory to fluoropyrimidines, oxaliplatin, or irinotecan. In December 2007, panitumumab was approved by the EMEA for use in patients with colorectal cancer who carry a normal, wild-type *K-RAS* gene.

* EGFR denotes epidermal growth factor receptor, EMEA European Medicines Evaluation Agency, FDA Food and Drug Administration, and ISEL Iressa Survival Evaluation in Lung Cancer.

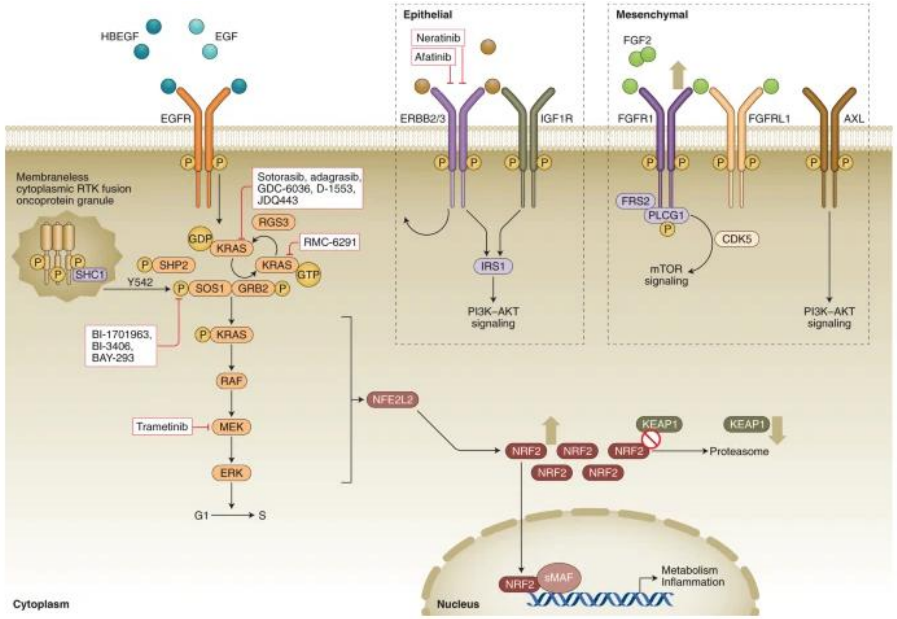
Ciardiello F, Tortora G. EGFR antagonists in cancer treatment. N Engl J Med. 2008 Mar 13;358(11):1160-74. doi: 10.1056/NEJMra0707704. Erratum in: N Engl J Med. 2009 Apr 9;360(15):1579. PMID: 18337605.

Applications of TKR drugs in cancer

As RAS protein play an important role in cancer development→ potential target for cancer therapy



Normal activation of EGFR- but activation of RAS pathway
Strategies for tackling cancer through
RAS protein



RAS-GDP-inactive → RAS-GTP-activate
Blocking GTP substrate

Limited efficacy- resistance development

Applications of TKR drugs in cancer

Do NOT study



Strategies for tackling cancer

RAS protein

Current direct KRAS-mutant inhibitors.

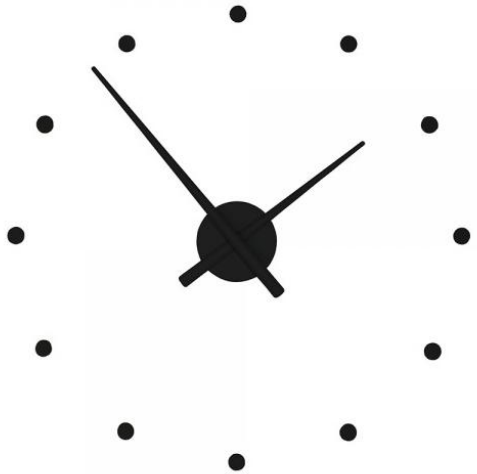
Drug	Alternate Names	Manufacturer	Target	Development Stage
ARS-853		Wellspring Biosciences	G12C	In vivo
ARS-1620		Wellspring Biosciences	G12C	In vivo
AMG510	Sotorasib	Amgen	G12C	FDA-approved
MRTX849	Adagrasib	Mirati therapeutics	G12C	FDA-approved
MRTX-EX185		Mirati therapeutics	G12D	In vivo
MRTX-1133		Mirati therapeutics	G12D	Phase ½ Clinical trials
ASP2453		Astellas Pharma Inc	G12C	In vivo
RMC-6291		Revolution Medicines	G12C	Phase 1 Clinical trial
RMC-6236		Revolution Medicines	G12C, G12D, G12V, G12X	In vivo (not published)
RMC-036		Revolution Medicines	G12D	In vivo (not published)
RMC-037		Revolution Medicines	G12D	In vivo (not published)
BBO-8520		Bridgebio Pharma	G12C	In vivo (not published)
ERAS-3490		Erasca	G12C	In vivo (not published)
JDQ443		Novartis	G12C	Phase 1–3 Clinical trials

Tria SM, Burge ME, Whitehall VLJ. The Therapeutic Landscape for *KRAS*-Mutated Colorectal Cancers. Cancers (Basel). 2023 Apr 19;15(8):2375. doi: 10.3390/cancers15082375. PMID: 37190303; PMCID: PMC10136970.

1st round of questions- Test TKR and cancer



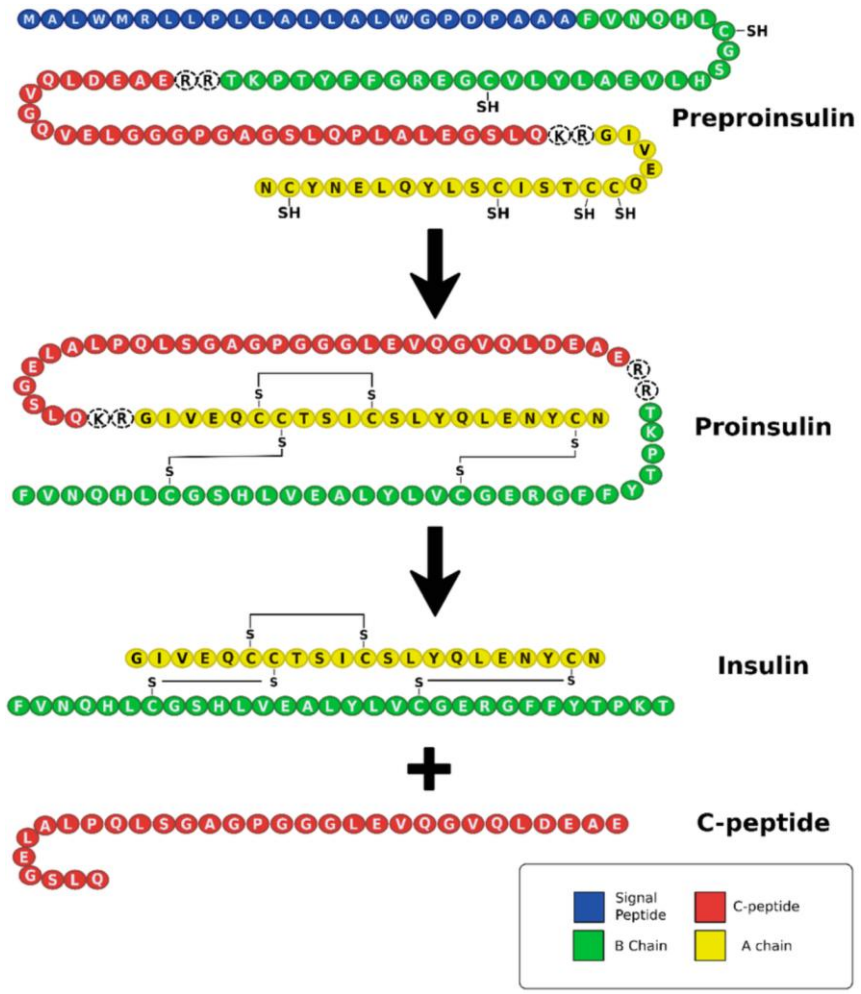
Break - 10 minutes



More about TKR : Insulin Receptor (INSR)

TKR : INSR family

Insulin: Polypeptide **hormone** coming from the proteolysis of **preproinsulin (prohormone).**





pancreas

Functions of Insulin:

Reduce glucose blood levels
(toxic high levels over the time)

Glucose storage for energy

Release in the blood stream

C-peptide reflects endogenous insulin secretion.

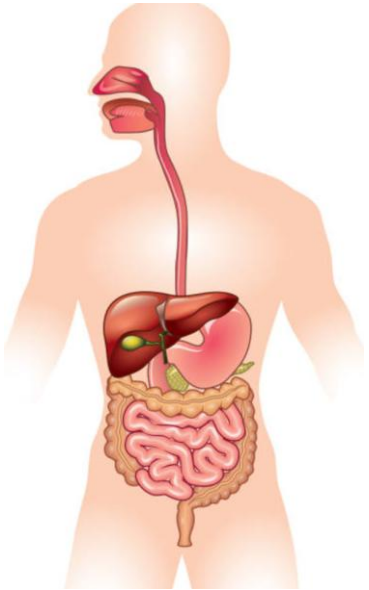
Preserved C-peptide levels indicate pancreatic insulin production, but do not establish whether it is sufficient or whether tissues respond appropriately.

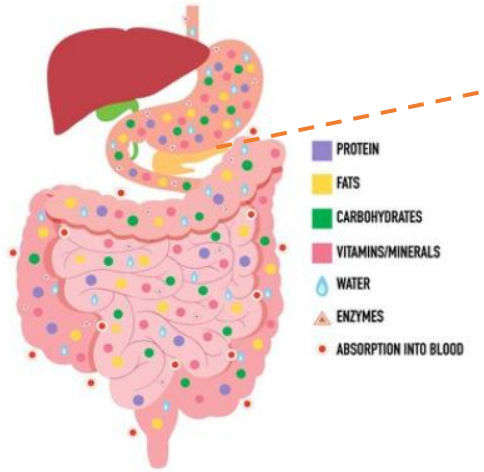
C-peptide alone cannot determine the type of diabetes.

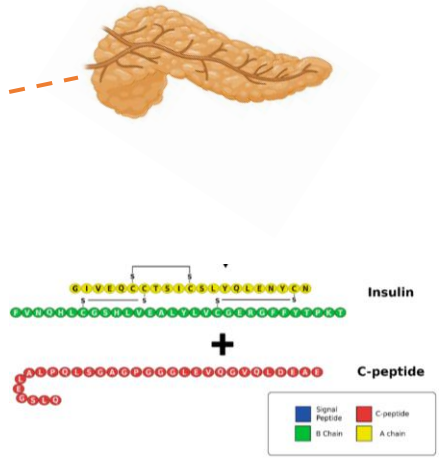
More about TKR : Insulin Receptor (INSR)

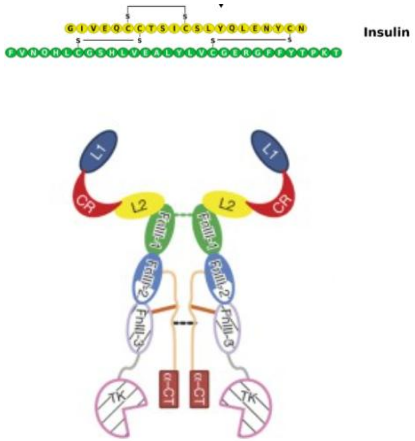
TKR : INSR family

Food intake - - - - - - **Digestion** - - - - - - **Insulin secretion** - - - - - - **Insulin binding to INSR**





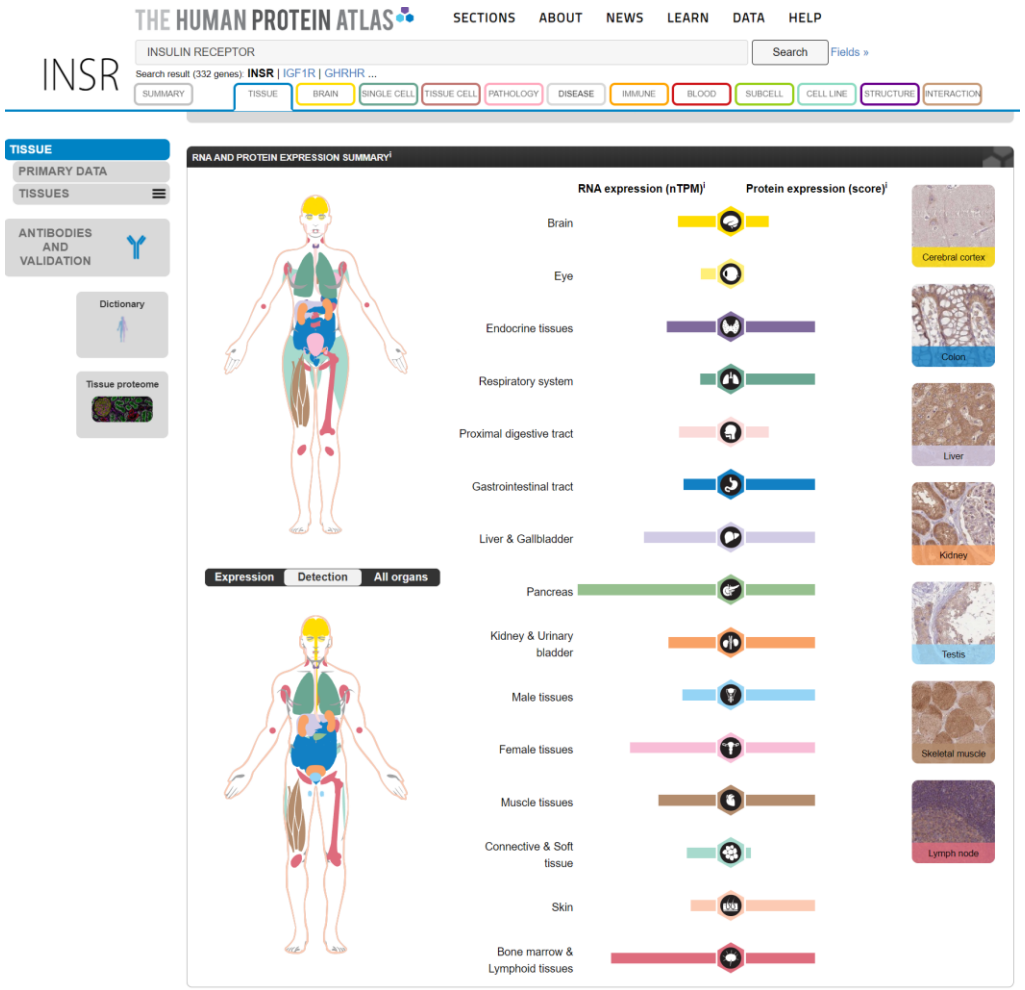




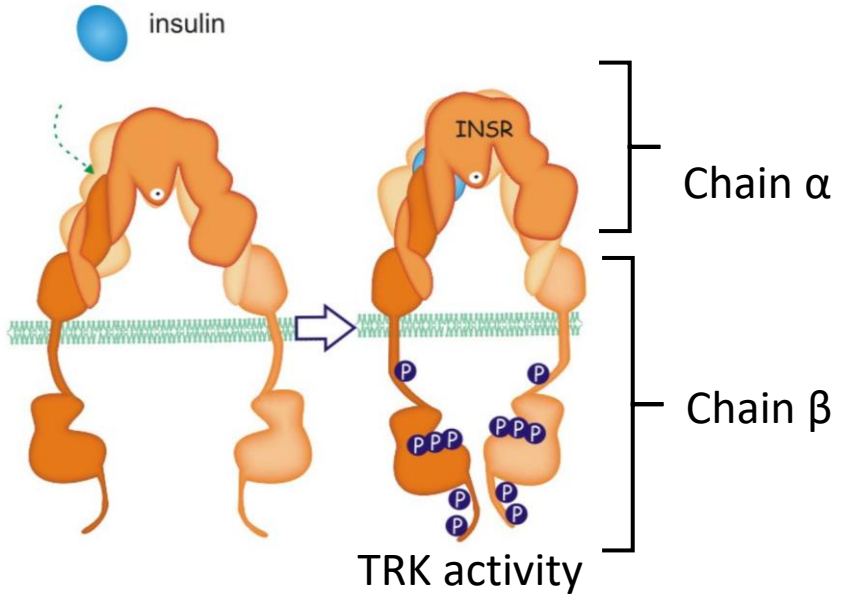
Liver
Adipose tissue
Skeletal Muscle
Brain

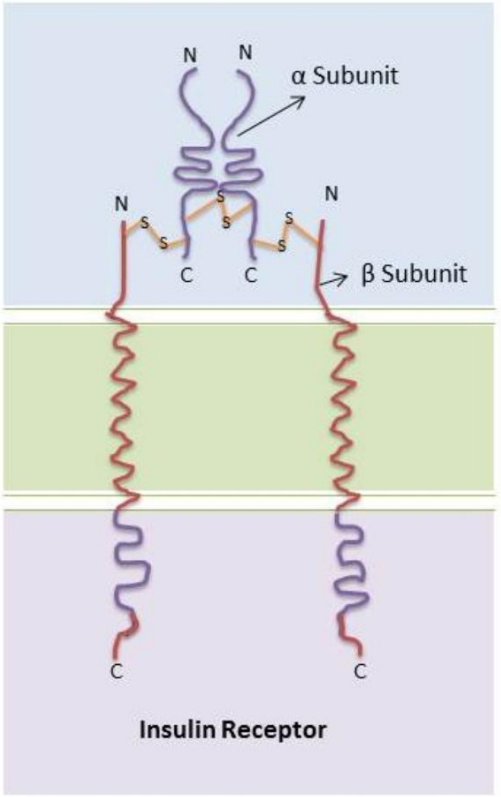
More about TKR : Insulin Receptor (INSR)

TKR : INSR family



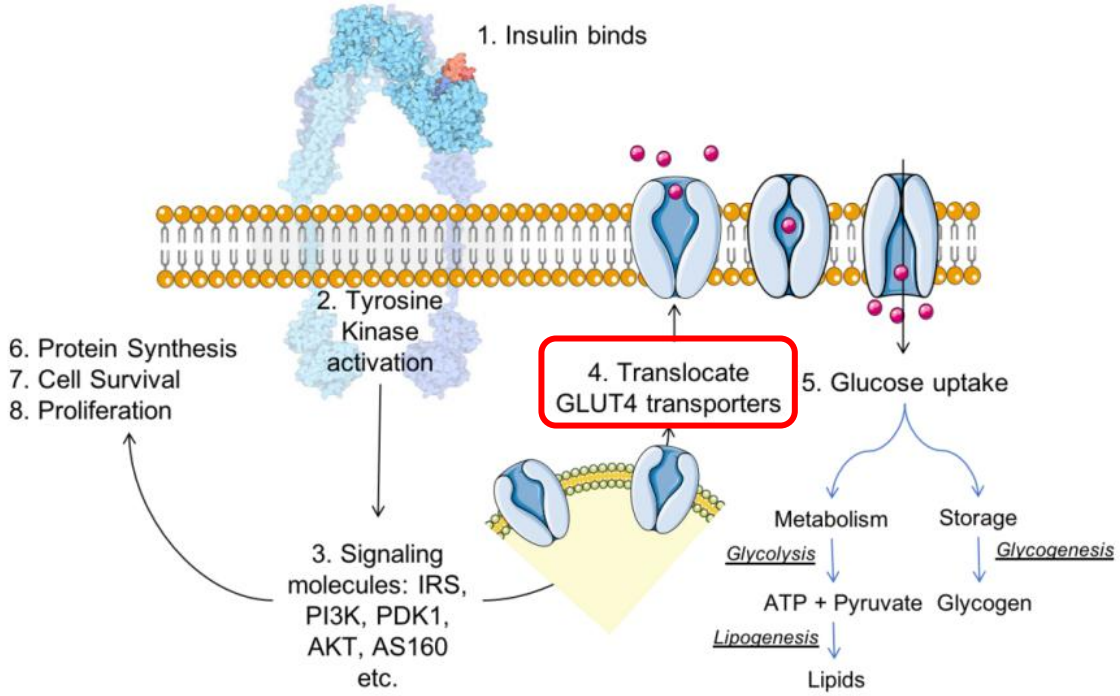
Structure:





More about TKR : Insulin Receptor (INSR)

TKR : INSR family



Insulin→ INSR→ **Glucose uptake**

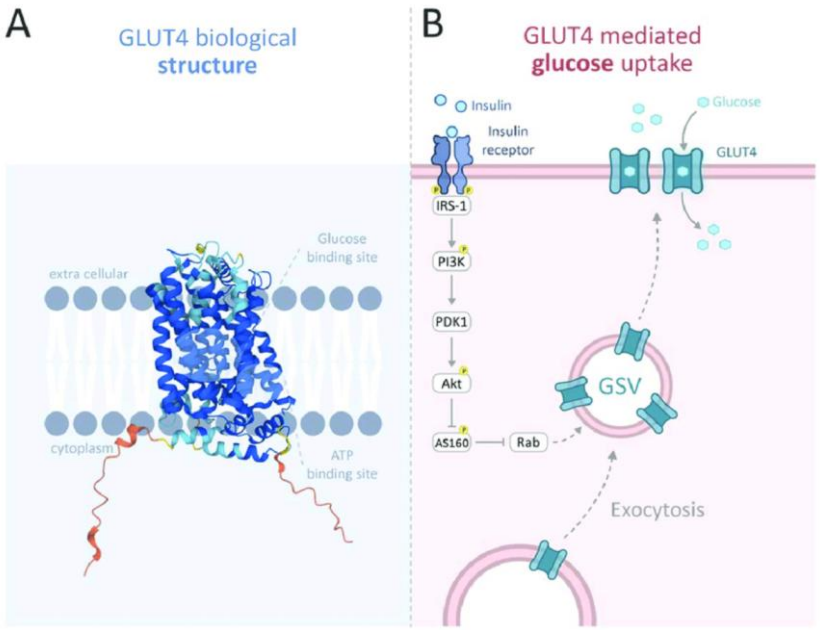
1. Metabolized for energy→ ATP production (Glycolysis) (Muscle)
2. Converted to glycogen for storage (Glycogenesis) (Liver/muscle)
3. Stored as triglycerides (Lipogenesis) (Adipose tissue)
4. Used for cell growth processes (Protein synthesis, cell survival and cell proliferation).

How does glucose uptake happens?

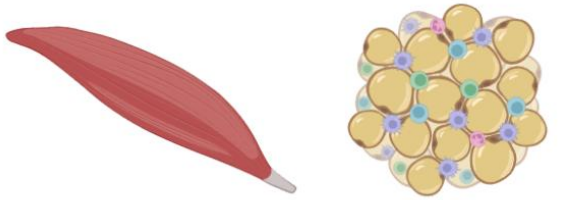
Traslocation of GLUT-4 transporters

More about TKR : Insulin Receptor (INSR)

TKR : INSR family- Traslocation GLUT-4 transporters



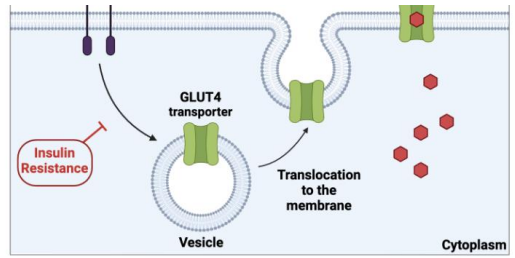
GLUT-4 transporter



Skeletal Muscle

Adipose tissue

GLUT-4 transporter

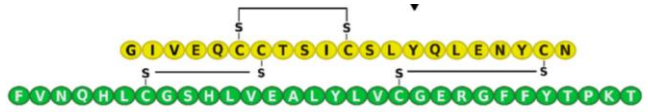


Cytoplasm
Mostly in endosomes/vesicles

The GLUT protein family (13 members) : membrane proteins form an aqueous pore across the cellular membrane through whichglucose/fructose/substrates can move.

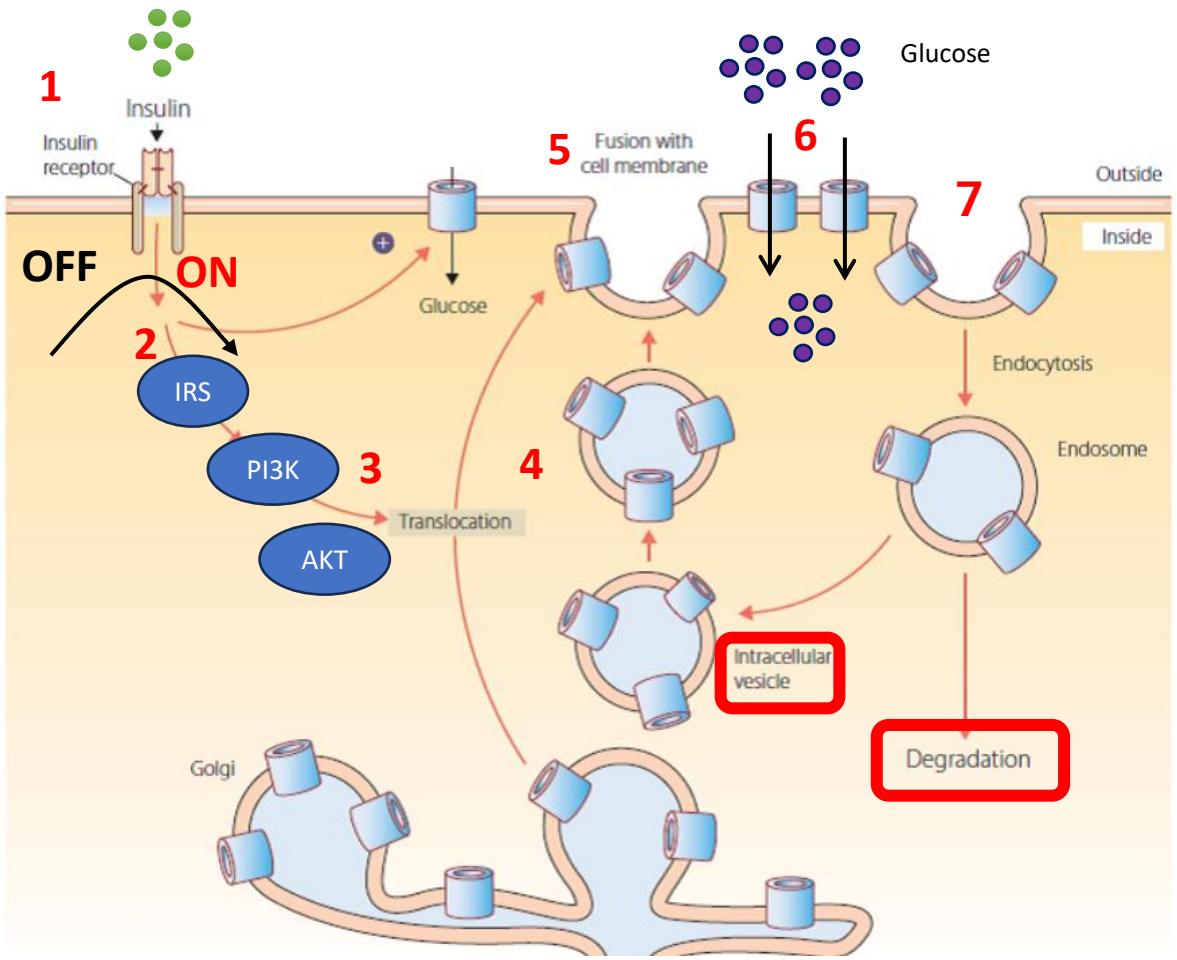
- **Class I (GLUTs 1–4) →glucose transporters**
- Class II (GLUTs 5, 7, 9 and 11)→fructose transporters
- Class III (GLUTs 6, 8, 10, 12 and HMIT1)→ several substrates.

What is the stimuly that made GLUT-4 transporters go from cytoplasm endosomes to cellular membrane??



More about TKR Insulin Receptor (INSR)

TKR : INSR family- Glucose storage



Glucose storage pathway

1- Ligand binds to membrane receptor / Insulin → INSR

2- Tyrosin kinase receptor activation/ Crossphosphorilation

3- Activation of IRS- PI3K-AKT signalling pathway

4-Traslocation of GLUT-4 transporters from cytoplasm vesicles to the membrane

5- GLUT-4 cellular membrane fusion

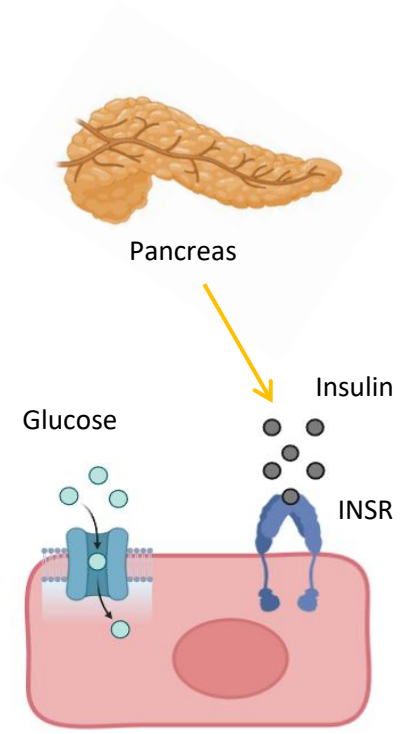
6- Glucose transport and storage

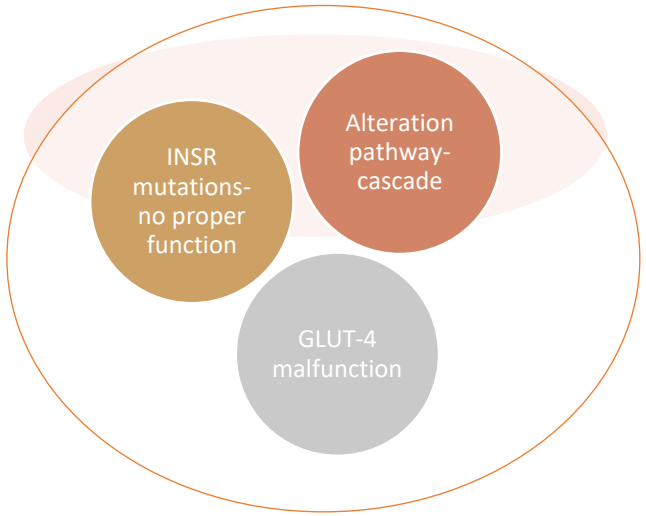
7- Decrease glucose blood levels→ GLUT-4 endocytosis

https://www.youtube.com/watch?v=MGMdKvZti1U

Medical relevance of INRS

No pathological conditions





Insulin resistance/prediabetes

Final stages: Diabetes

National Institute of Diabetes and Digestive and Kidney Diseases

What is insulin resistance?

Insulin resistance is when cells in your muscles, fat, and liver don't respond well to insulin and can't easily take up glucose from your blood. As a result, your pancreas makes more insulin to help glucose enter your cells. As long as your pancreas can make enough insulin to overcome your cells' weak response to insulin, your blood glucose levels will stay in the healthy range.

Clinically, the term "insulin resistance" implies that higher-than-normal concentrations of insulin are required to maintain normoglycemia.

Who is more likely to develop insulin resistance or prediabetes?

- overweight or obesity
- age 45 or older
- a parent, brother, or sister with diabetes
- African American, Alaska Native, American Indian, Asian American, Hispanic/Latino, Native Hawaiian, or Pacific Islander American ethnicity
- physical inactivity
- health conditions such as high blood pressure and abnormal cholesterol levels

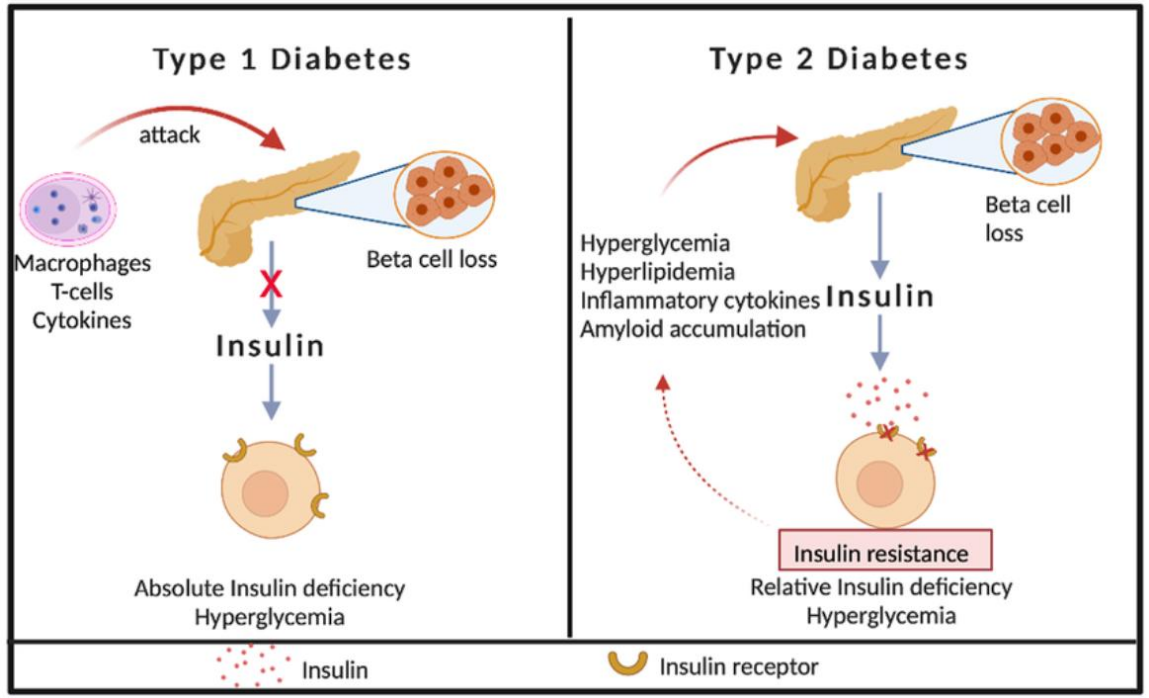
Medical relevance of TKR in diabetes

What is diabetes?

World Heath Organization (WHO) definition:

"Diabetes is a chronic disease that occurs either when the pancreas does not produce enough insulin or when the body cannot effectively use the insulin it produces"

Autoimmune disease



Metabolic disease
Immune disease??
Ag production

Applications of TKR drugs in diabetes

Do NOT study



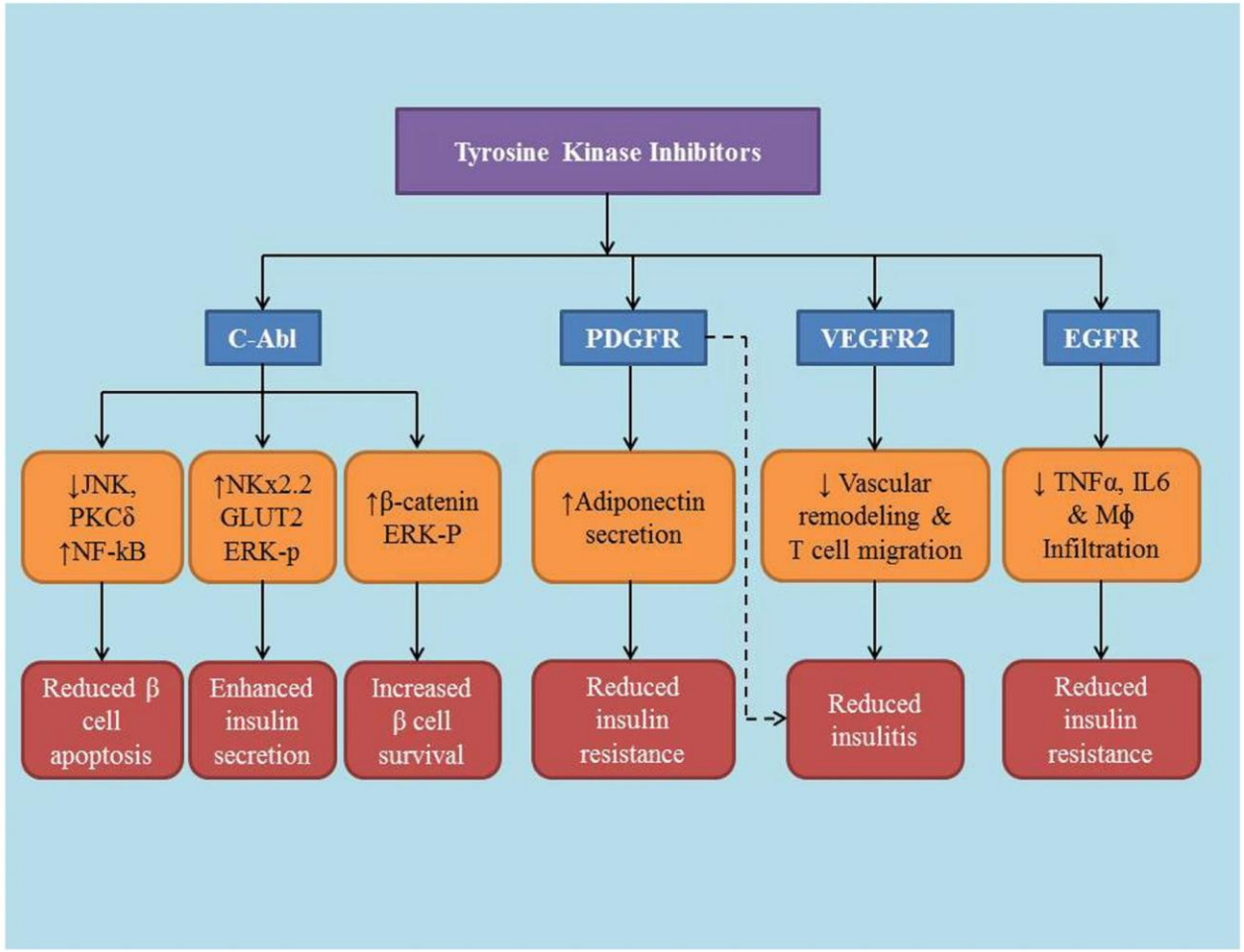
T1D: insulin

T2D most commom drugs:
Metformin.
SGLT2 inhibitors: empagliflozin and dapagliflozin.
GLP-1 receptor agonists: semaglutide, dulaglutide, and liraglutide.
Dual GIP/GLP-1 receptor agonist: tirzepatide.
Sulfonylureas: glipizide, glimepiride, and gliclazide.
DPP-4 inhibitors: sitagliptin and linagliptin.
Thiazolidinediones: pioglitazone.
Insulin

TKR inhibitors drugs

They are not used because they would impair glucose uptake, leading to hyperglycemia.

- Principal application→cancer treatment.
  - Positive indirect effect on diabetes.



Gefitinib (Lung cancer)

Finishing clinical case TKR, 2nd round of diabetesquestions, diabetes clinical case





Thank you!!!

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